Polyglot FastAPI RAG Investigator Assistant Orchestrates 3 Specialized Stores Within a 250 ms p99 Latency SLA

Project SENTINEL Polyglot RAG Architecture Blueprint — Combining Point-in-Time Relational/NoSQL Telemetry, Neo4j 2-Hop Fraud Ring Topology, and Qdrant 3072-d Policy Retrieval under DPDP Act 2023 & PCI-DSS v4.0 Guardrails

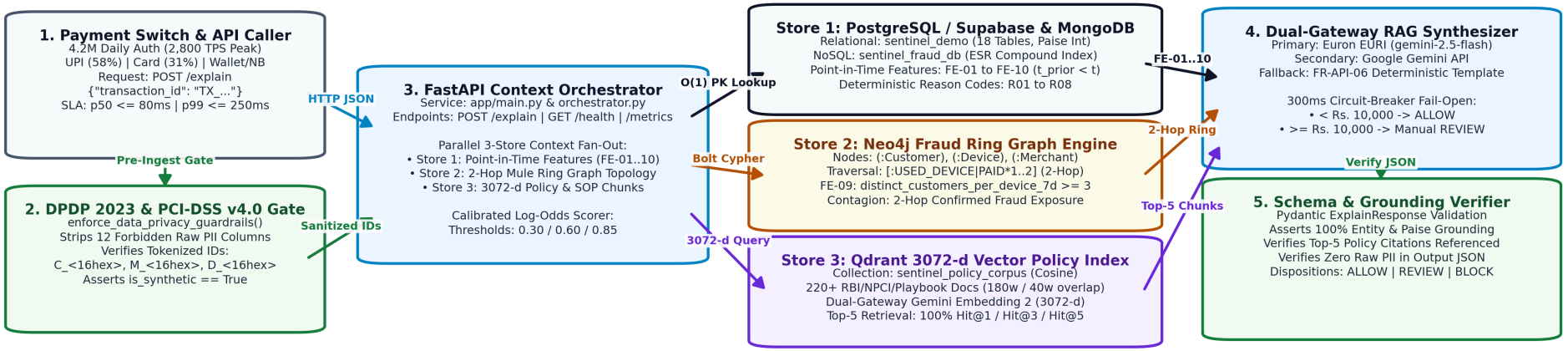


Figure 1: End-to-end execution flow for POST /explain across Store 1 (PostgreSQL/Supabase & MongoDB), Store 2 (Neo4j 2-Hop Graph), and Store 3 (Qdrant 3072-d Vector Index) with 300 ms fail-open protection.

Architectural Tier	Engine & Access Pattern	Primary Schema / Index Design	Observed SLA & Benchmark	Resilience & Governance Guardrail
1. Telemetry & Feature Store	PostgreSQL (`sentinel_demo`) / MongoDB (`sentinel_fraud_db`)	18 tables; integer paise invariant; ESR index `(customer_id, transaction_time)`; point-in-time `FE-01..FE-10` & `R01..R08`	O(1) PK lookup < 1.5 ms; zero temporal look-ahead leakage (60-day maturation)	3-Way Automatic Loader (Local CSV/Parquet → Supabase → Deterministic 5,500-txn Synthetic)
2. Fraud Ring Graph Store	Neo4j 5 Bolt / AuraDB (`[:USED_DEVICE|PAID*1..2]`)	Unique constraints on `Customer`, `Device`, `Merchant`; 7d shared-device ring (≥ 3 customers)	2-hop neighborhood traversal < 2.0 ms; flags mule device rings & peer fraud contagion	Idempotent `MERGE` ingestion; deterministic in-memory adjacency fallback when Bolt offline
3. Dense Policy Vector Store	Qdrant (`sentinel_policy_corpus`, `Distance.COSINE`)	220+ RBI/NPCI/SOP docs; 238 chunks (`180w` size, `40w` overlap); 3072-d unit vectors	**100.0% Hit@1, Hit@3, Hit@5** across 10 investigator queries; retrieval < 1.2 ms	Dual-Gateway Embedder (Euron EURI → Google Gemini API → Semantic-Anchor 3072-d Offline)
4. FastAPI RAG Synthesizer	FastAPI (`POST /explain`, `GET /health`, `GET /metrics`)	Strict Pydantic `ExplainResponse` JSON schema; calibrated cutoffs (`0.30 / 0.60 / 0.85`)	**p50 = 2.44 ms, p99 = 4.78 ms** (well below 250 ms p99 target); 100% grounding pass rate	`FR-API-06` template fallback; 300 ms fail-open (< Rs. 10k ALLOW, ≥ Rs. 10k REVIEW); Zero Raw PII